COMP9418 – Advanced Topics in Statistical Machine Learning

W8 – Undirected Graphical Models

(aka Markov Random Fields)

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September 17, 2018

Admin: Plan for the following weeks

- 24/09/2018 - 30/09/2018 (Teaching recess, no teaching or tutorials)

- Week 10: NSW Public holiday (Labour day, 01/10/2018): No face-to-face lecture this week but a video will be recorded and posted on the course website. Everything else as usual.

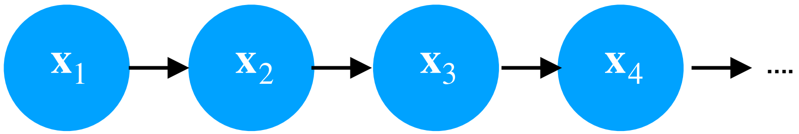
Acknowledgements

Material derived from:

- [Murphy, MLaPP, 2012] Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, 2012
  - Most figures taken from this book
- [Bishop, PRML, 2006] Pattern Recognition and Machine Learning, Christopher Bishop, 2006
  - Some figures taken from this book

Recap

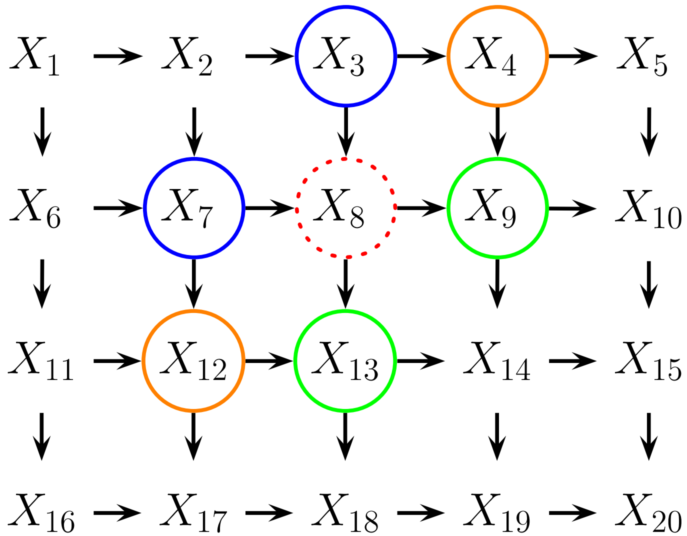
- (Hidden) Markov Models for modelling sequential data.
- Directed Graphical Models for modelling conditional (in)dependency assumptions



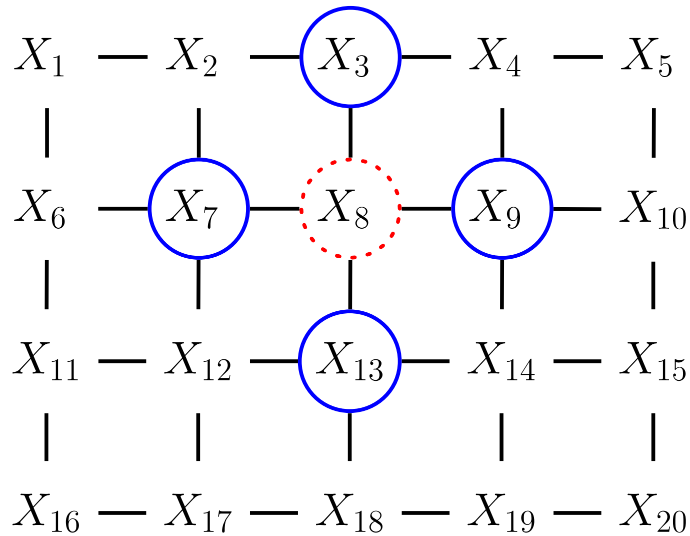
$$p(\mathbf{x}_{1:t}) = p(\mathbf{x}_1) \prod_{t=1}^{T} p(\mathbf{x}_t | \mathbf{x}_{t-1}).$$

Graphical Models for Images

Directed model



Undirected model



- Choosing the directionality of edges may be unnatural
- What is the Markov blanket of X_8?
- UGMs are much more natural for problems such as image analysis and spatial statistics

Aims (1)

This lecture will allow you to understand undirected graphical models and apply them to machine learning problems. Following it you should be able to:

- Determine conditional independence assumptions made by a given undirected graphical model
- Differentiate between directed graphical models and undirected graphical models
- Represent joint distributions as normalised products of potentials and as linear models in the log space
- Carry out parameter estimation in Markov random fields (MRFs) via gradient-based optimisation of the likelihood

Aims (2)

- Formulate and apply discriminative models based on conditional random fields (CRFs)
- Differentiate between generative models and discriminative models
- Understand and solve inference problems required in parameter estimation for MRFs and CRFs
- Determine suitable applications for MRFs

Outline

I. Conditional Independence in UGMs

II. Directed Graphical Models (DGMs) vs Undirected Graphical Models (UGMs)

III. Representing Probability Distributions with Markov Random Fields (MRFs)

IV. Parameter Learning in MRFs

V. Conditional Random Fields (CRFs)

I. Conditional Independence in UGMs

Recall the Definition of Markov Blanket

In general:

Markov Blanket of a Node

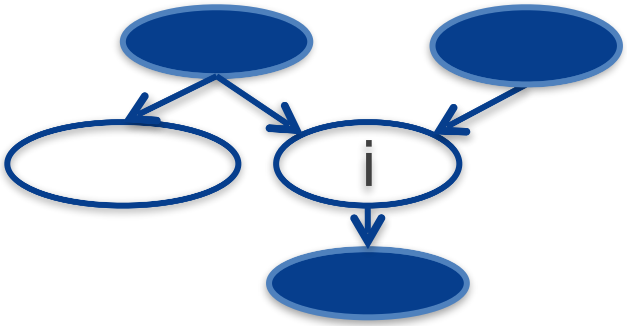
The smallest set of nodes that renders it conditionally independent of all other nodes

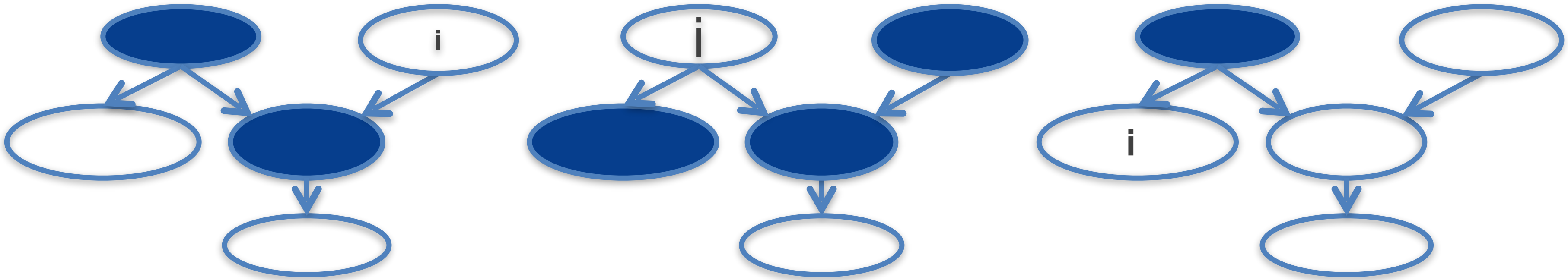
This applies to both directed graphical models (DGMs) and undirected graphical models (UGMs)

Markov Blanket in DGMs

All what wee need to guess node *i*

- Parents of node *i*
- Children of node *i*
- Parents of children of node *i*



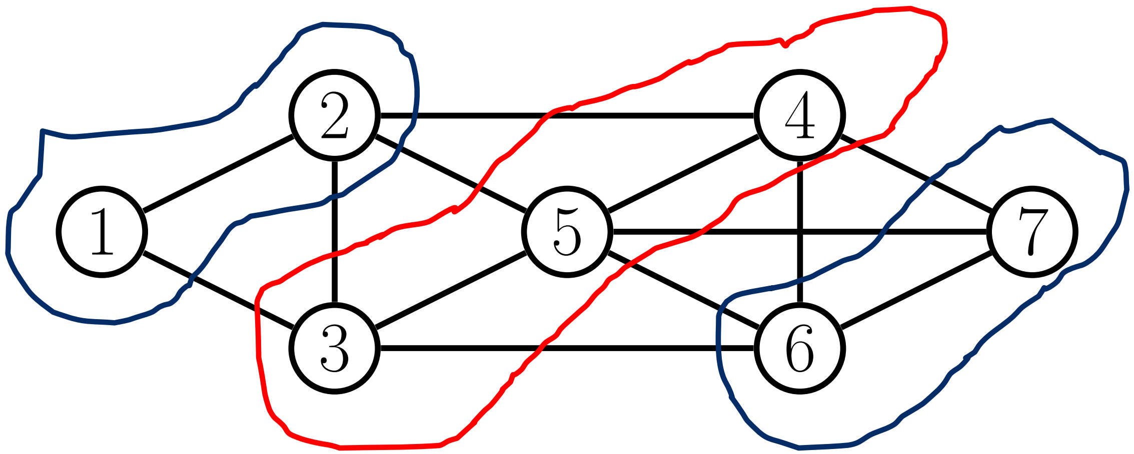


Conditional Independence Properties in UGMs

P1. Global Markov Property

We say: $\mathbf{x}_A \perp\!\!\!\perp \mathbf{x}_B \mid \mathbf{x}_C$ iff C separates A from B in the graph G

When removing all the nodes in C, if there are no paths connecting any node in A to any node in B then the CI property holds.



$$\{1,2\} \perp\!\!\!\perp \{6,7\} \mid \{3,4,5\}$$

Markov Blanket in UGMs

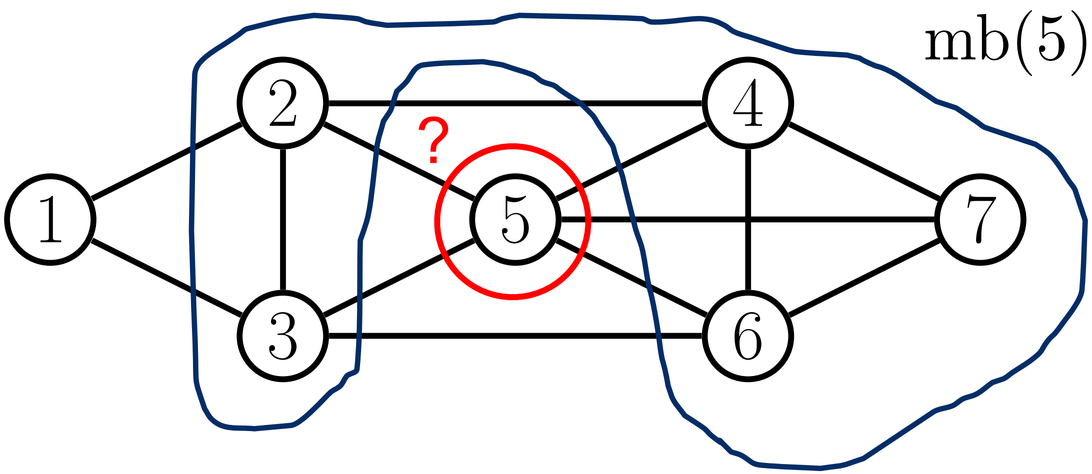
Formally, we denote the Markov blanket of note *t* with: $\mathrm{mb}(t)$

And it satisfies: $t \perp\!\!\!\perp \mathcal{V} \backslash \mathrm{cl}(t) \mid \mathrm{mb}(t)$

Where $\mathrm{cl}(t) \stackrel{\text{def}}{=} \{t\} \cup \mathrm{mb}(t)$ is the *closure* of *t*

P2. Undirected local Markov property

In a UGM, a node's Markov blanket is its set of immediate neighbours



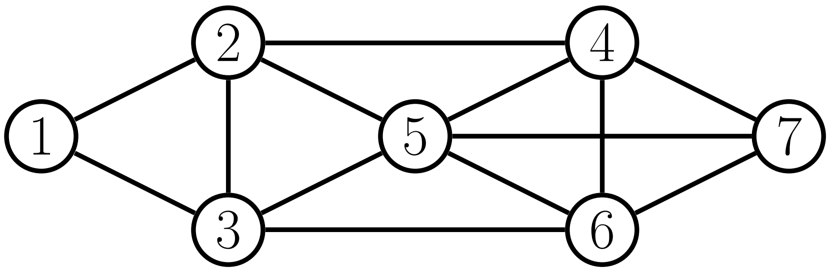
$\mathrm{mb}(5) \stackrel{\text{def}}{=} \{2, 3, 4, 6, 7\}$

More CI Properties in UGMs

P3. Pairwise Markov property

Two nodes are conditionally independent given the rest if there is no direct edge between them

$$s \perp\!\!\!\perp t \mid \mathcal{V} \backslash \{s, t\} \iff G_{st} = 0$$



P3 (P) $1 \perp\!\!\!\perp 7 \mid \text{rest}$

P2 (L) $1 \perp\!\!\!\perp \text{rest} \mid \{2, 3\}$

P1 (G) $\{1\ 2\} \perp\!\!\!\perp \{6, 7\} \mid \{3, 4, 5\}$

Local, Global and Pairwise are the Same

- Global implies local, which implies pairwise
- More interestingly, assuming a positive density, p(**x**) > 0, pairwise implies global
- Hence, all these Markov properties are the same!

$$\mathrm{G} \Longrightarrow \mathrm{L} \Longrightarrow \mathrm{P}$$

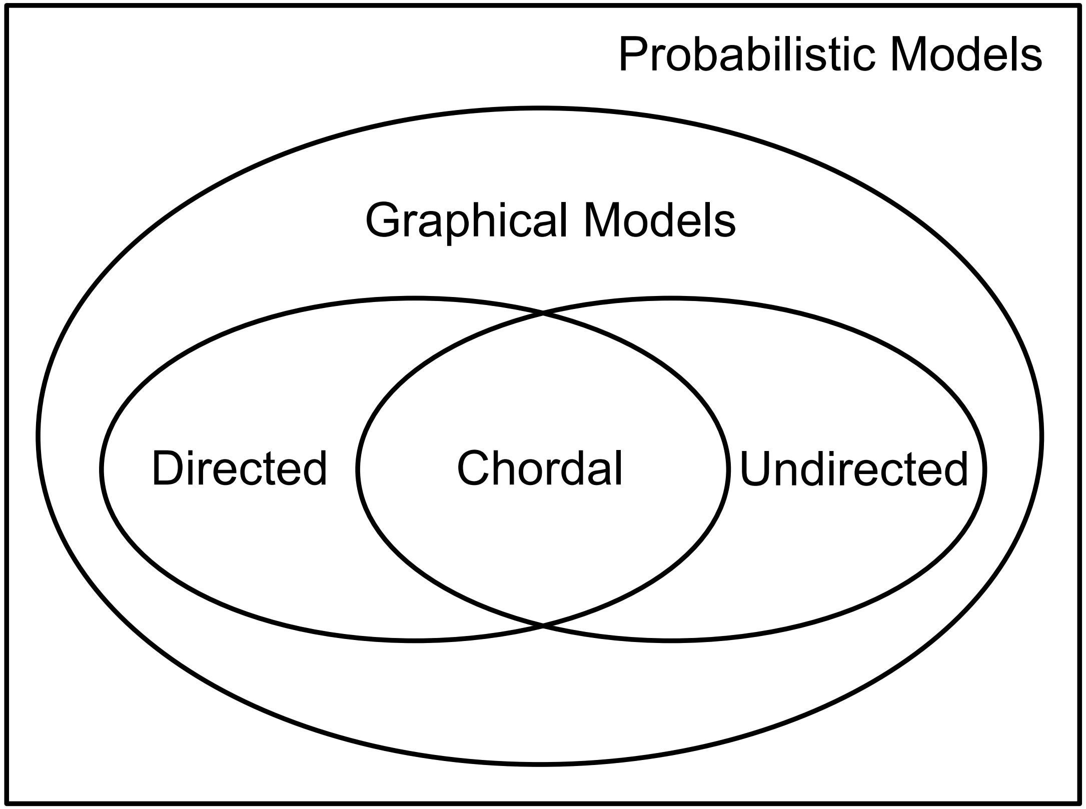
- Such pairwise CI statements can be used to construct a graph from which global CI statements can be extracted

$$p(y_i \,|\, \text{rest}) = p(y_i \,|\, \text{neighbours of } i)$$

II. Directed Graphical Models (DGMs) vs Undirected Graphical Models (UGMs)

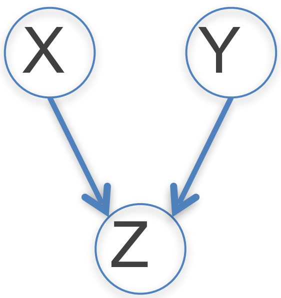
Which one is more expressive DGM or UGM?

- **Perfect Map:** The graph can represent all (and only) the CI properties of the distribution
  - DGMs and UGMs are perfect maps for different sets of distributions
  - Neither is more powerful as a representational language



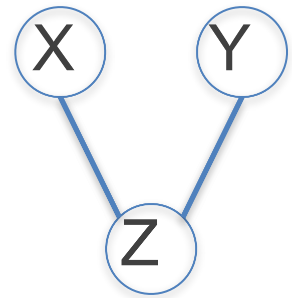
DGMs vs UGMs

- **Example (1):** CI relationships perfectly modelled by a DGM but not a UGM



$$X \perp\!\!\!\perp Y$$

$$X \not\perp\!\!\!\perp Y \mid Z$$

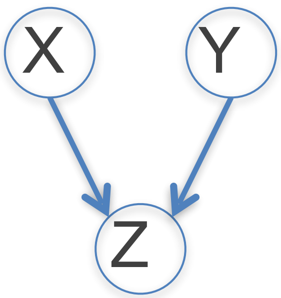


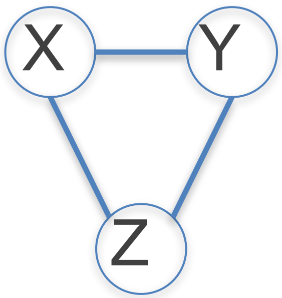
$$X \not\perp\!\!\!\perp Y$$

$$X \perp\!\!\!\perp Y \mid Z$$

DGMs vs UGMs

- **Example (2):** CI relationships perfectly modelled by a DGM but not a UGM





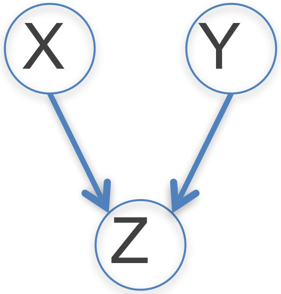
$$X \perp\!\!\!\perp Y \qquad\qquad X \not\perp\!\!\!\perp Y$$

$$X \not\perp\!\!\!\perp Y \mid Z \qquad\qquad X \not\perp\!\!\!\perp Y \mid Z$$

- There is no UGM that can precisely represent all and only the two CI statements encoded by a v-structure
- Non-monotonicity of DGMs: Conditioning on extra variables can eliminate conditional independences

Non-monotonicity of DGMs

- **Non-monotonicity of DGMs: Conditioning on extra variables can eliminate conditional independences**



$$X \perp\!\!\!\perp Y$$

$$X \not\!\perp\!\!\!\perp Y \mid Z$$

DGM

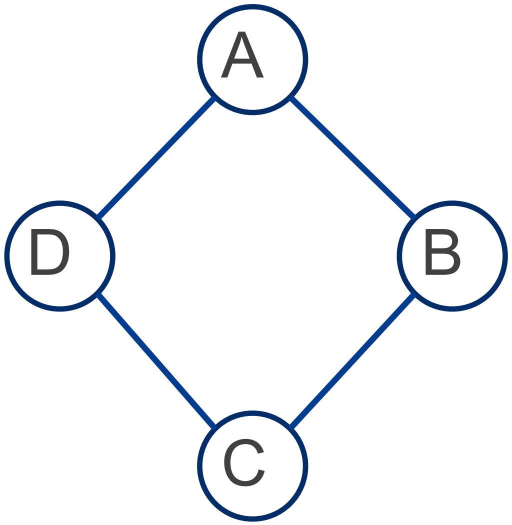
$$X \perp\!\!\!\perp Y \mid C$$

$$X \perp\!\!\!\perp Y \mid \{C, \ldots\}$$

UGM

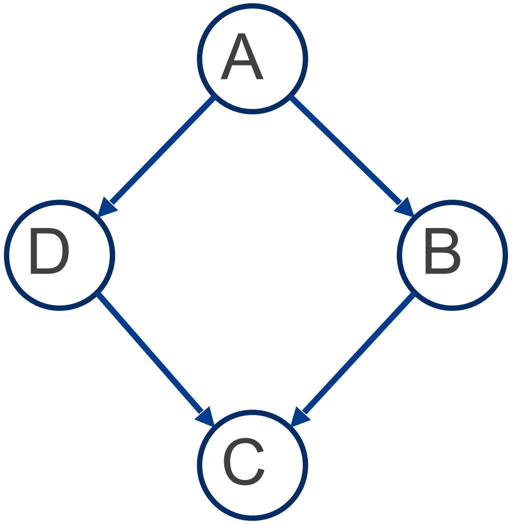
DGMs vs UGMs

- **Example (3):** CI relationships perfectly modelled by a UGM but not a DGM



$$A \perp\!\!\!\perp C \mid B, D$$

$$B \not\!\perp\!\!\!\perp D \mid A$$

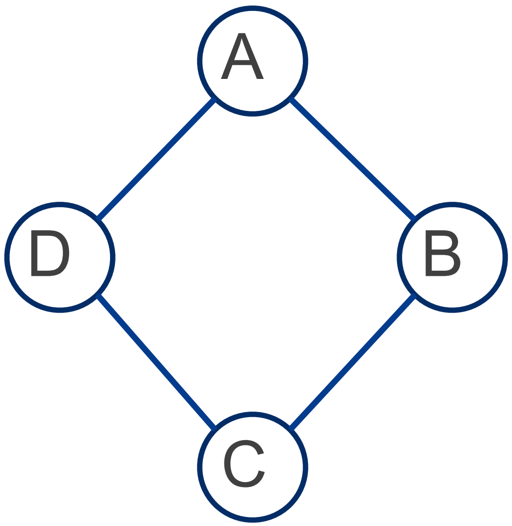


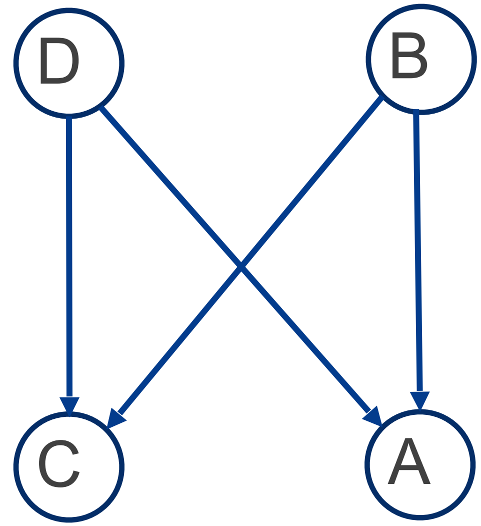
$$A \perp\!\!\!\perp C \mid B, D$$

$$B \perp\!\!\!\perp D \mid A$$

DGMs vs UGMs

- **Example (4):** CI relationships perfectly modelled by a UGM but not a DGM





$$A \perp\!\!\!\perp C \mid B, D$$
$$B \not\perp\!\!\!\perp D$$

$$A \perp\!\!\!\perp C \mid B, D$$
$$B \perp\!\!\!\perp D$$

- There is no DGM that can precisely represent all and only the two CI statements encoded by this UGM

III. Representing Probability Distributions with Markov Random Fields (MRFs)

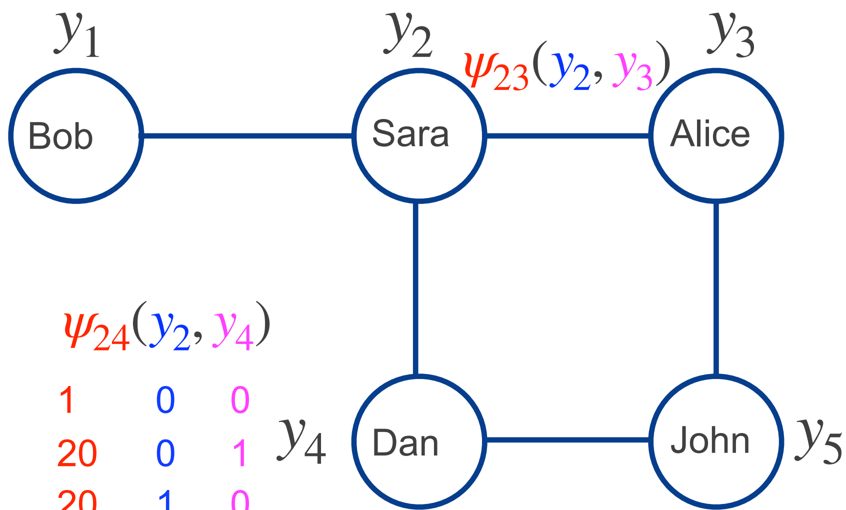
Parameterisation of MRFs: Example

y_i : 0: there is no misconception
1: there is a misconception

potential function(factor)

$\psi_{12}(y_1, y_2)$

40	0	0
1	0	1
1	1	0
10	1	1



$\psi_{23}(y_2, y_3)$

$\psi_{35}(y_3, y_5)$

200	0	0
1	0	1
1	1	0
200	1	1

$\psi_{24}(y_2, y_4)$

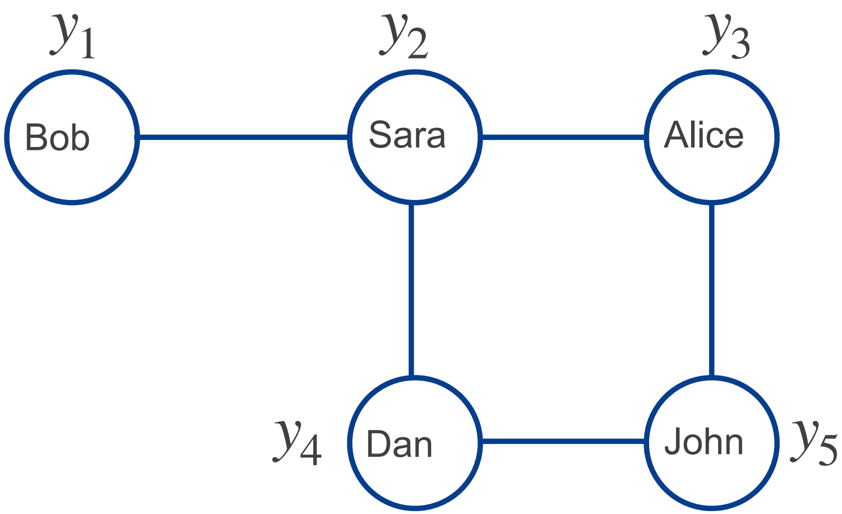
1	0	0
20	0	1
20	1	0
1	1	1

$\psi_{45}(y_4, y_5)$

$$p(y_1, y_2, y_3, y_4, y_5) \propto \psi_{12}(y_1, y_2)\psi_{23}(y_2, y_3)\psi_{35}(y_3, y_5)\psi_{45}(y_4, y_5)\psi_{24}(y_2, y_4)$$

https://www.youtube.com/watch?v=KWNtPHVf2VQ

Parameterisation of MRFs: Example



y_1, y_2, y_3, y_4, y_5	$\psi_{12}(y_1, y_2) \ldots \psi_{24}(y_2, y_4)$
0 0 0 0 0	200
1 0 0 0 0	5
⋮	⋮
1 1 1 1 1	100
	Z

partition function → Z

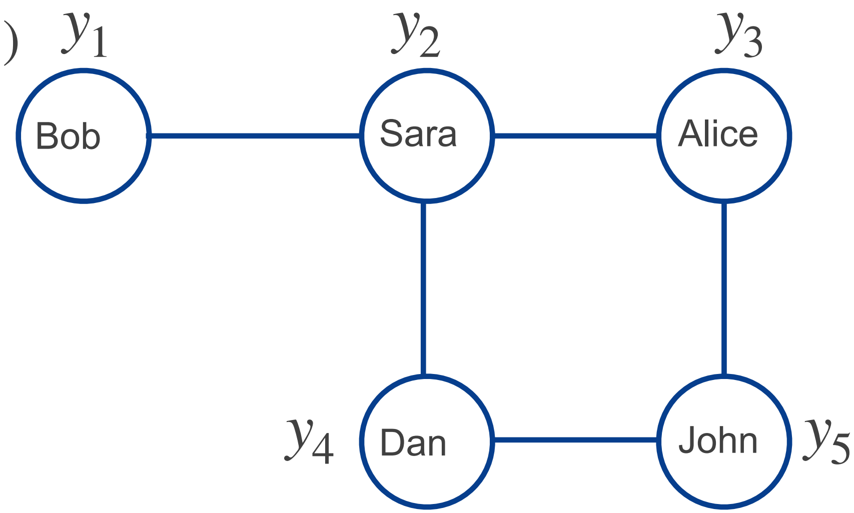
$$p(y_1, y_2, y_3, y_4, y_5) = \frac{\psi_{12}(y_1, y_2)\psi_{23}(y_2, y_3)\psi_{35}(y_3, y_5)\psi_{45}(y_4, y_5)\psi_{24}(y_2, y_4)}{\sum_{y_1, y_2, y_3, y_4, y_5} \psi_{12}(y_1, y_2)\psi_{23}(y_2, y_3)\psi_{35}(y_3, y_5)\psi_{45}(y_4, y_5)\psi_{24}(y_2, y_4)}$$

https://www.youtube.com/watch?v=KWNtPHVf2VQ

Parameterisation of MRFs: Example

y_i : 0: there is no misconception
1: there is a misconception

$\psi_{12}(y_1, y_2 \mid \theta)$		
θ_1	0	0
θ_2	0	1
θ_3	1	0
θ_4	1	1



$$\psi_c(y_c \mid \theta_c)$$

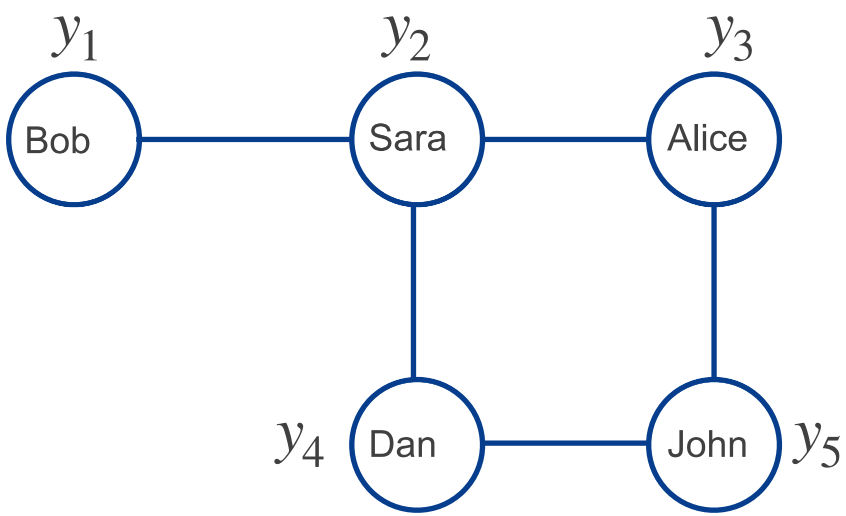
$$Z(\theta)$$

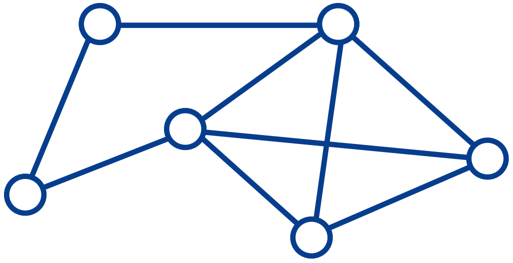
Parameterisation of MRF

- But can we represent any UGM using ψ functions?

$$p(y_i \,|\, \text{rest}) = p(y_i \,|\, \text{neighbours of } i)$$

y_i : 0: there is no misconception
1: there is a misconception





The Hammersley-Clifford theorem

Hammersley-Clifford (1971):

A positive distribution p(y) > 0 *satisfies the CI properties of an undirected graph* G *iff* p *can be represented as a product of factors, one per maximal clique, i.e.,*

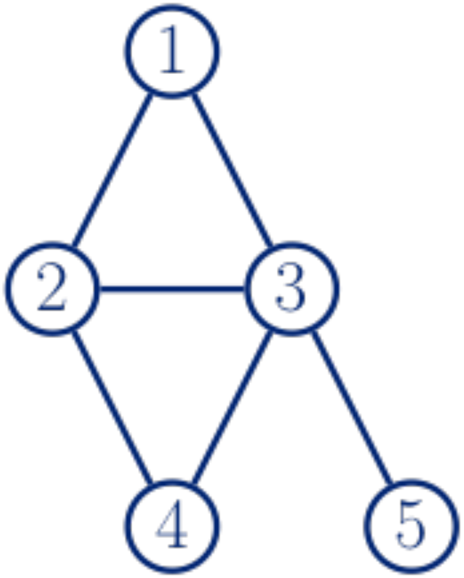
$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$$

Where $\mathcal{C}$ *is the set of all maximal cliques of* G *and* $Z(\boldsymbol{\theta})$ *is the partition function:*

$$Z(\boldsymbol{\theta}) \stackrel{\text{def}}{=} \sum_{\mathbf{y}} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$$

Ensures normalisation

MRF Example



$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})}\psi_{123}(y_1, y_2, y_3)\psi_{234}(y_2, y_3, y_4)\psi_{35}(y_3, y_5)$$

where $$Z(\boldsymbol{\theta}) = \sum_{y_1,\dots,y_5} \psi_{123}(y_1, y_2, y_3)\psi_{234}(y_2, y_3, y_4)\psi_{35}(y_3, y_5)$$

Parameterisation of MRFs

- Associate *potential functions* with each maximal clique in the graph
- Denote potential function for clique c by $\psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$
  - θ_c are the parameters of the potential
- Potential Function: Any non-negative function of its arguments
- Define joint distribution to be proportional to the product of these potentials:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$$

UGMs and Statistical Physics

- Consider the Gibbs distribution:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp\left(-\sum_c E(\mathbf{y}_c|\boldsymbol{\theta}_c) \right)$$

  – $E(\mathbf{y}_c) > 0$: Energy associated with variables $\mathbf{y}_c$

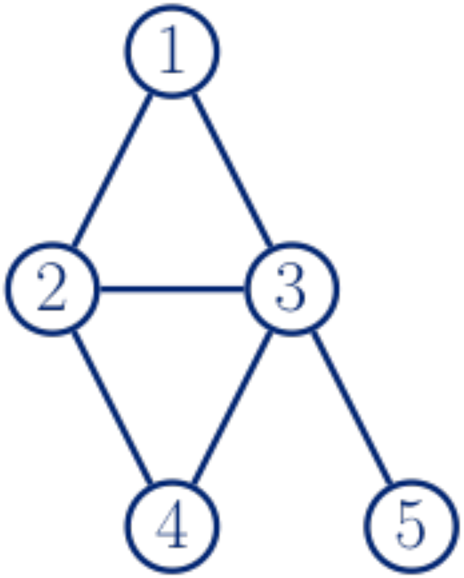
- We can convert this to a UGM by simply making:

$$\psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c) = \exp(-E(\mathbf{y}_c|\boldsymbol{\theta}_c))$$

  – High probability states correspond to low-energy configuration
  – Usually known as energy-based models

Pairwise MRFs

- Parameterisation can be restricted to the edges of the graph



$$p(\mathbf{y}|\boldsymbol{\theta}) \propto \psi_{12}(y_1, y_2)\psi_{13}(y_1, y_3)\psi_{23}(y_2, y_3)\psi_{24}(y_2, y_4)\psi_{34}(y_3, y_4)\psi_{35}(y_3, y_5)$$

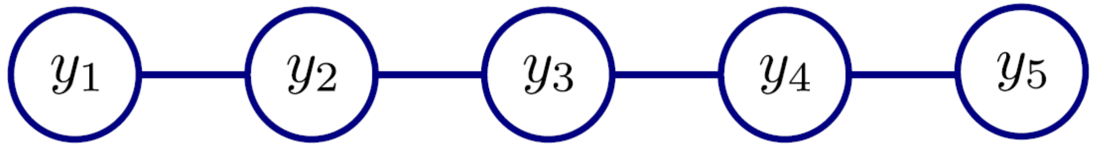
$$= \prod_{s \sim t} \psi_{st}(y_s, y_t)$$

Product over neighbours

- Widely used due to simplicity (although not as general)

Pairwise MRFs

- Example: A smooth chain



$$p(y_1 \ldots y_5) \propto \psi_{12}(y_1, y_2)\psi_{23}(y_2, y_3)\psi_{34}(y_3, y_4)\psi_{45}(y_4, y_5)$$

Consider the case where $y_i \in \{0,1\}$

$$\psi_{mn}(1,1) = 1 \qquad \psi_{mn}(1,0) = 0.1$$

$$\psi_{mn}(0,1) = 0.1 \qquad \psi_{mn}(0,0) = 1$$

$y_{1\ldots5}$	$Pr(y_{1\ldots5})$	$y_{1\ldots5}$	$Pr(y_{1\ldots5})$	$y_{1\ldots5}$	$Pr(y_{1\ldots5})$	$y_{1\ldots5}$	$Pr(y_{1\ldots5})$
00000	0.09877	01000	0.02469	10000	0.04938	11000	0.04938
00001	0.04938	01001	0.01235	10001	0.02469	11001	0.02469
00010	0.02469	01010	0.00617	10010	0.01235	11010	0.01235
00011	0.04938	01011	0.01235	10011	0.02469	11011	0.02469
00100	0.02469	01100	0.02469	10100	0.01235	11100	0.04938
00101	0.01235	01101	0.01235	10101	0.00617	11101	0.02469
00110	0.02469	01110	0.02469	10110	0.01235	11110	0.04938
00111	0.04938	01111	0.04938	10111	0.02469	11111	0.09877

https://inst.eecs.berkeley.edu/~cs280/sp15/lectures/14.pdf

Representing Potential Functions

Discrete Variables

- Tables of non-negative numbers
  - Similarly to CPTs, although potentials are not probabilities
  - They represent compatibility between different assignments
- More generally, we can have a log-linear model

$$\log \psi_c(\mathbf{y}_c) \stackrel{\text{def}}{=} \boldsymbol{\phi}_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c$$

$$\psi_c(\mathbf{y}_c) = e^{\phi_c(\mathbf{y}_c)^T \theta_c}$$

Tabular MRFs as log-linear models

- Consider pairwise MRFs, where for each edge we have a K2-dimensional feature vector:

$$\boldsymbol{\phi}_{st}(y_s, y_t) = (\ldots, \mathbb{I}(y_s = j, y_t = k), \ldots)$$

Example:

$$\phi_{st}(y_s = 0, y_t = 0) = [1,0,0,0]$$
$$\phi_{st}(y_s = 1, y_t = 0) = [0,1,0,0]$$
$$\phi_{st}(y_s = 0, y_t = 1) = [0,0,1,0]$$
$$\phi_{st}(y_s = 1, y_t = 1) = [0,0,0,1]$$

- Having a weight for each feature, we can write the potential as:

$$\psi_{st}(y_s, y_t) = \exp(\boldsymbol{\theta}_{st}^T \boldsymbol{\phi}_{st})$$

Example:

$$\log \psi_{st}(y_s, y_t) = \theta_1\phi_1 + \theta_2\phi_2 + \theta_3\phi_3 + \theta_4\phi_4$$

- Log-linear form is more general

Representing Potential Functions

Discrete Variables

- Tables of non-negative numbers
  - Similarly to CPTs, although potentials are not probabilities
  - They represent compatibility between different assignments
- More generally, we can have a log-linear model

$$\log \psi_c(\mathbf{y}_c) \stackrel{\text{def}}{=} \boldsymbol{\phi}_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c$$

- Hence the log probability is given by:

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$$

$$\log p(\mathbf{y}|\boldsymbol{\theta}) = \sum_c \boldsymbol{\phi}_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c - \log Z(\boldsymbol{\theta})$$

  - Also known as maximum entropy model

IV. Parameter Learning in MRFs

Learning In MRFs (1)

- Parameter estimation computationally expensive
  - Point estimation (e.g. MLE) instead of full Bayesian inference

- Consider an MRF in log linear form:

$$\log p(\mathbf{y}|\boldsymbol{\theta}) = \sum_c \boldsymbol{\phi}_c(\mathbf{y}_c)^T \boldsymbol{\theta}_c - \log Z(\boldsymbol{\theta})$$

- The average log-likelihood is given by:

$$\mathscr{L}(\theta) = \frac{1}{N}\sum_{n=1}^{N} \log p(\mathbf{y}^{(n)}\,|\,\theta) = \frac{1}{N}\sum_{n=1}^{N}\left[\sum_c \phi_c(\mathbf{y}_c^{(n)})^T\theta_c - \log Z(\theta))\right]$$

  - Concave in θ
  - Unique global maximum using gradient-based optimisation

Learning In MRFs (2)

- For a particular clique *c*, the derivatives of the weights are:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_c} = \frac{1}{N} \sum_{n=1}^{N} \left[\phi_c(\mathbf{y}_c^{(n)}) - \frac{\partial}{\partial \boldsymbol{\theta}_c} \log Z(\boldsymbol{\theta}) \right]$$

- It is possible to show that the derivative of the log partition function wrt θ_c is:

$$\frac{\partial \log Z(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}_c} = \sum_{\mathbf{y}_c} \phi_c(\mathbf{y}_c) p(\mathbf{y}_c | \boldsymbol{\theta})$$

$$= \mathbb{E}_{p(\mathbf{y}_c | \boldsymbol{\theta})} [\phi_c(\mathbf{y}_c)]$$

– The expectation of the c feature under the model

Learning In MRFs (3)

Hence:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_c} = \boxed{\frac{1}{N}\sum_{n=1}^{N} \phi_c(\mathbf{y}_c^{(n)})} - \boxed{\mathbb{E}_{p(\mathbf{y}_c|\boldsymbol{\theta})}[\phi_c(\mathbf{y}_c)]}$$

- Empirical expectation

$$\mathbb{E}_{p_{\text{emp}}}[\boldsymbol{\phi}_c(\mathbf{y})]$$

- **y** fixed to observed values
  - Clamped term

- Model expectation
- **y** is free
  - Unclamped term or contrastive term

- At the optimum we have: $\mathbb{E}_{p_{\text{emp}}}[\phi_c(\mathbf{y}_c)] = \mathbb{E}_{p(\mathbf{y}_c|\boldsymbol{\theta})}[\phi_c(\mathbf{y}_c)]$
  - Moment matching
- Computing the unclamped term requires inference in the model once per gradient step → computationally demanding

Iterative Proportional Fitting (IPF, 1)

- Fixed-point iteration
  - Assume we are interested to find fixed-point of *g:* $x = g(x)$
  - One way is to do the following iteration: $x^{t+1} = g(x^t)$
  - Given some conditions on *g*, x^t converges to the fixed point of *g.*

- The aim is to have $p_{\text{emp}}(\mathbf{y}_c) = p(\mathbf{y}_c \mid \psi_c)$.
  - We define:

$$g(\psi_c) = \frac{p_{\text{emp}}(\mathbf{y}_c)}{p(\mathbf{y}_c \mid \psi_c)} \psi_c(\mathbf{y}_c)$$

  - At fixed point:

$$g(\psi_c) = \frac{p_{\text{emp}}(\mathbf{y}_c)}{p(\mathbf{y}_c \mid \psi_c)} \psi_c(\mathbf{y}_c) = \psi_c(\mathbf{y}_c) \rightarrow \quad p_{\text{emp}}(\mathbf{y}_c) = p(\mathbf{y}_c \mid \psi_c)$$

  - Iterate as follows:

$$\psi_c^{t+1}(\mathbf{y}_c) = \psi_c^t(\mathbf{y}_c) \frac{p_{\text{emp}}(\mathbf{y}_c)}{p(\mathbf{y}_c \mid \psi^t)}$$

Iterative Proportional Fitting (IPF, 2)

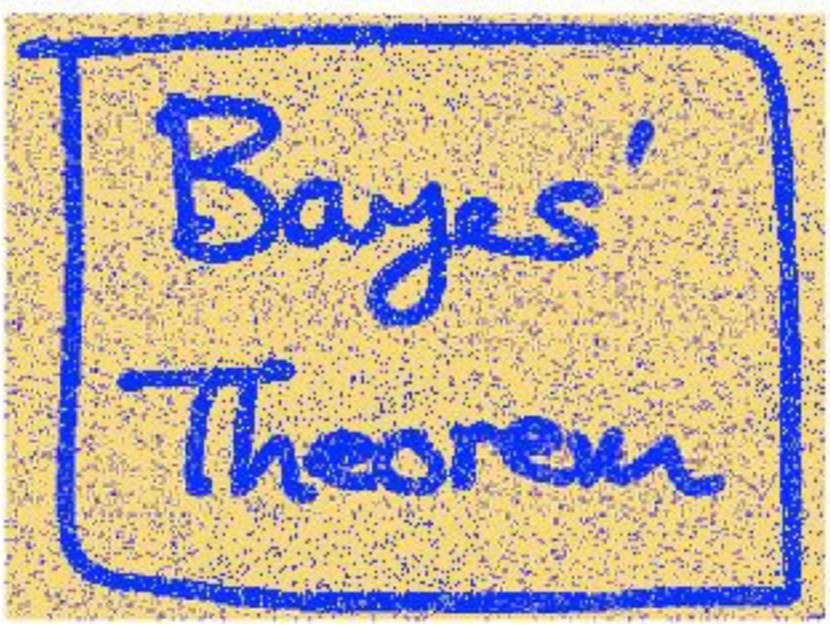
1. initialise $\psi_c = 1$ for $c = 1, \ldots, C$;
2. **repeat**
3. **for** $c = 1 : C$ **do**
4. $\quad p_c = p(\mathbf{y}_c | \boldsymbol{\psi})$;
5. $\quad \hat{p}_c = p_{\text{emp}}(\mathbf{y}_c)$;
6. $\quad \psi_c = \psi_c * \dfrac{\hat{p}_c}{p_c}$;
7. **until** converged.

- Guaranteed to converge to global optimum
- Need to compute required marginals under the model
  - Can use the JTA
- Convergence can be slow

Illustration: Image De-Noising (1)

(Original slide from Bishop, PRML, 2006)

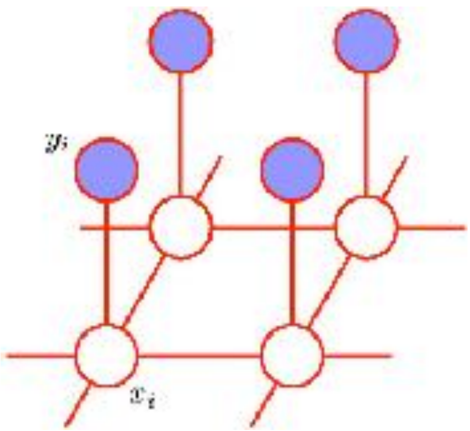
Original Image



Noisy Image

Illustration: Image De-Noising (2)

(Original slide from Bishop, PRML, 2006)



orignal image $x_i \in \{-1,1\}$

observed images $y_i \in \{-1,1\}$

strong correlation between x_i and y_i

bias for one type of pixels

$$E(\mathbf{x},\mathbf{y}) = -\eta \sum_i x_i y_i - \beta \sum_{\{i,j\}} x_i x_j + h \sum_i x_i$$

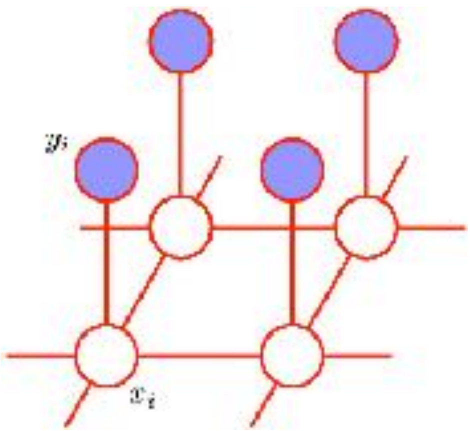
to be minimised

strong correlation between x_i and its neighbours

$$p(\mathbf{x},\mathbf{y}) = \frac{1}{Z} \exp\{-E(\mathbf{x},\mathbf{y})\}$$

Illustration: Image De-Noising (2)

(Original slide from Bishop, PRML, 2006)



orignal image $x_i \in \{-1,1\}$

observed images $y_i \in \{-1,1\}$

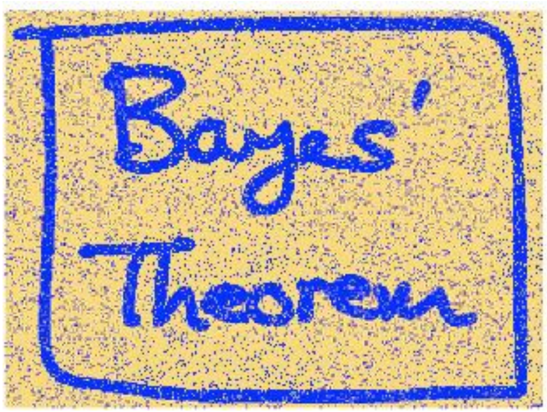
$$E(\mathbf{x},\mathbf{y}) = -\eta \sum_i x_i y_i - \beta \sum_{\{i,j\}} x_i x_j + h \sum_i x_i$$

Goal: find $\mathbf{x}$ that minimise $E(\mathbf{x},\mathbf{y})$

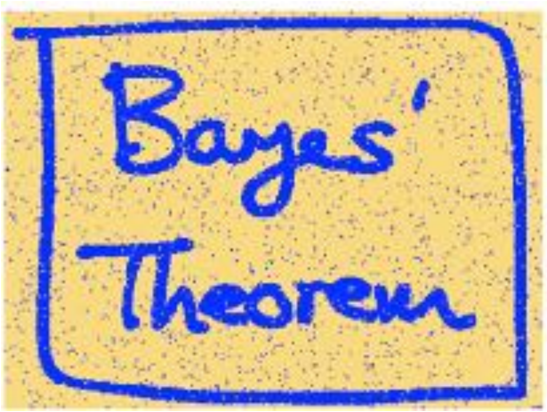
Illustration: Image De-Noising (3)

(Original slide from Bishop, PRML, 2006)

- Iterated conditional modes (or ICM)
  - Try $x_i = 1$ and $x_i = -1$ and see which one decreases $E(\mathbf{x}, \mathbf{y})$
  - Converges to local maximum of probability



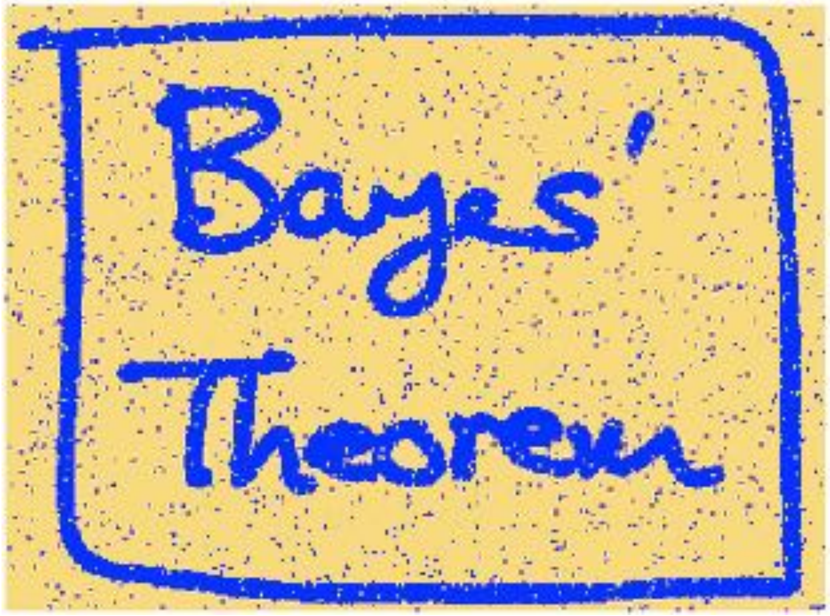
Noisy Image



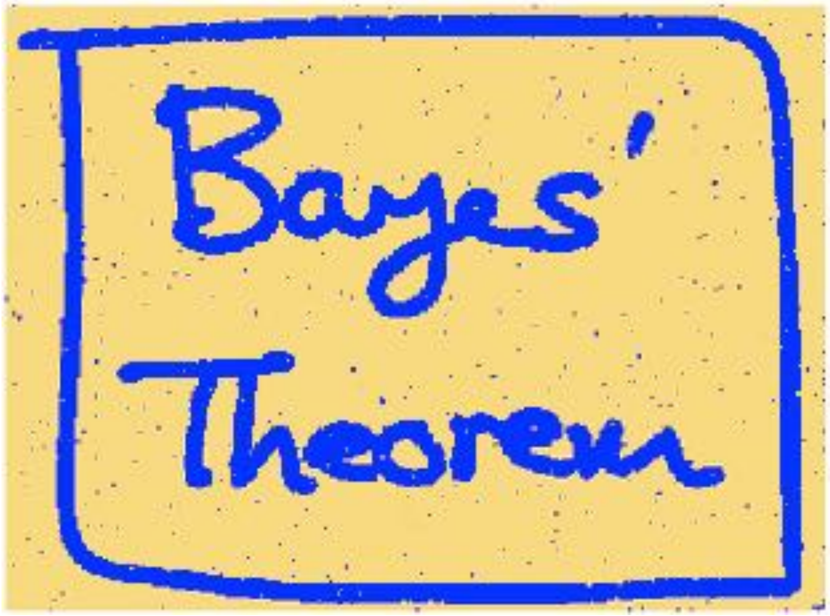
Restored Image (ICM)

Illustration: Image De-Noising (4)

(Original slide from Bishop, PRML, 2006)



Restored Image
(ICM)



Restored Image
(Graph cuts)

V. Conditional Random Fields (CRFs)

Conditional Random Fields (CRFs)

Handwriting recognition

X:

Y:

b	r	a	c	e

sequence of letters

$$p(\mathbf{y}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\boldsymbol{\theta}_c)$$

$\psi_{ij}(y_i, y_j)$ language bigram model

$$p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) = \frac{1}{Z(\mathbf{x}, \boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\mathbf{x}, \boldsymbol{\theta})$$

Conditional Random Fields (CRFs)

- *A conditional random field (CRF) is a MRF where all the clique potentials are conditioned on input features.*

$$p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) = \frac{1}{Z(\mathbf{x}, \boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\mathbf{x}, \boldsymbol{\theta})$$

Conditional Random Fields (CRFs)

Example

Handwriting recognition

X:

Y:

b	r	a	c	e

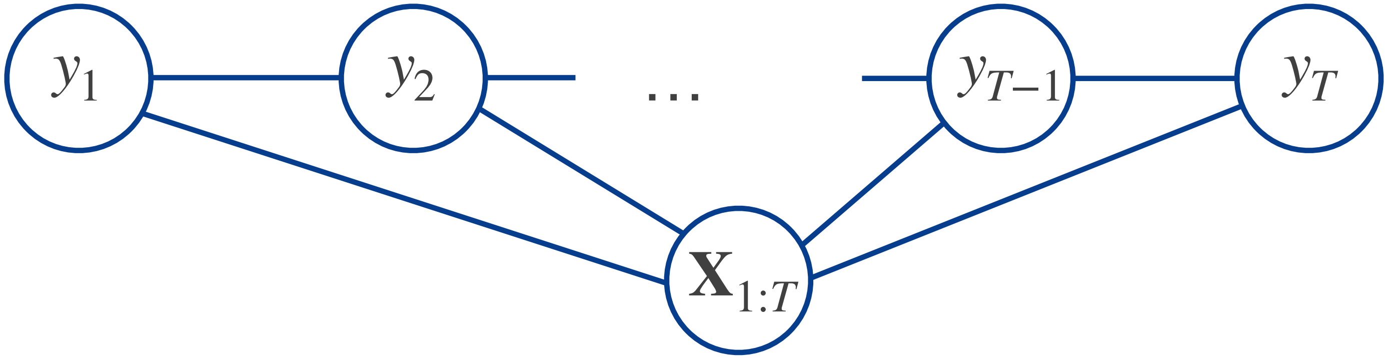
Step1: Modelling conditional probabilities:

$$p(\mathbf{y}|\mathbf{x}, \boldsymbol{\theta}) = \frac{1}{Z(\mathbf{x}, \boldsymbol{\theta})} \prod_{c \in \mathcal{C}} \psi_c(\mathbf{y}_c|\mathbf{x}, \boldsymbol{\theta})$$

Step2: finding most probable y:

$$\mathbf{y}^* = \arg\max_{y} p(\mathbf{y} \,|\, \mathbf{x})$$

Chain-structured CRF



- Much focus on the *linear-chain* CRF:

$$p(\mathbf{y} \mid \mathbf{x}, \theta) = \frac{1}{Z(\mathbf{x}, \theta)} \prod_{t=1}^{T} \psi(y_t, y_{t+1} \mid \mathbf{x}, \theta)$$

  – With unary and binary features:

$$p(\mathbf{y} \mid \mathbf{x}, \theta) = \frac{1}{Z(\mathbf{x}, \theta)} \prod_{t=1}^{T} \psi(y_t \mid \mathbf{x}, \theta) \prod_{t=1}^{T-1} \psi(y_t, y_{t+1} \mid \mathbf{x}, \theta)$$

Conditional Random Fields (CRFs)

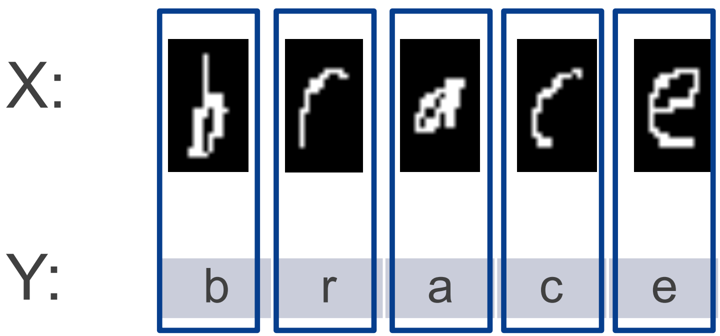
Example: Handwriting recognition

$$p(\mathbf{y} \mid \mathbf{x}, \theta) = \frac{1}{Z(\mathbf{x}, \theta)} \prod_{t=1}^{T} \psi(y_t \mid \mathbf{x}, \theta) \prod_{t=1}^{T-1} \psi(y_t, y_{t+1} \mid \mathbf{x}, \theta)$$

Handwriting recognition

local classifier: if x_i is y_i

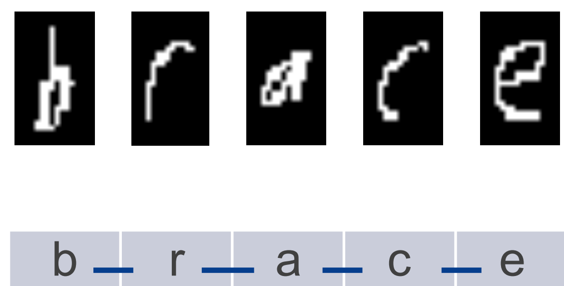
$\psi_i(y_i \mid x_i)$



X

language bigram model

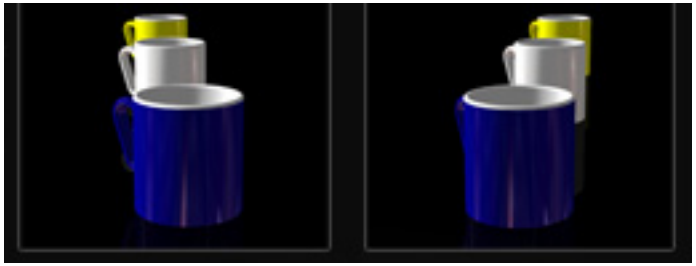
$\psi_{ij}(y_i, y_j)$



Conditional Random Fields (CRFs)

Example: Stereo vision

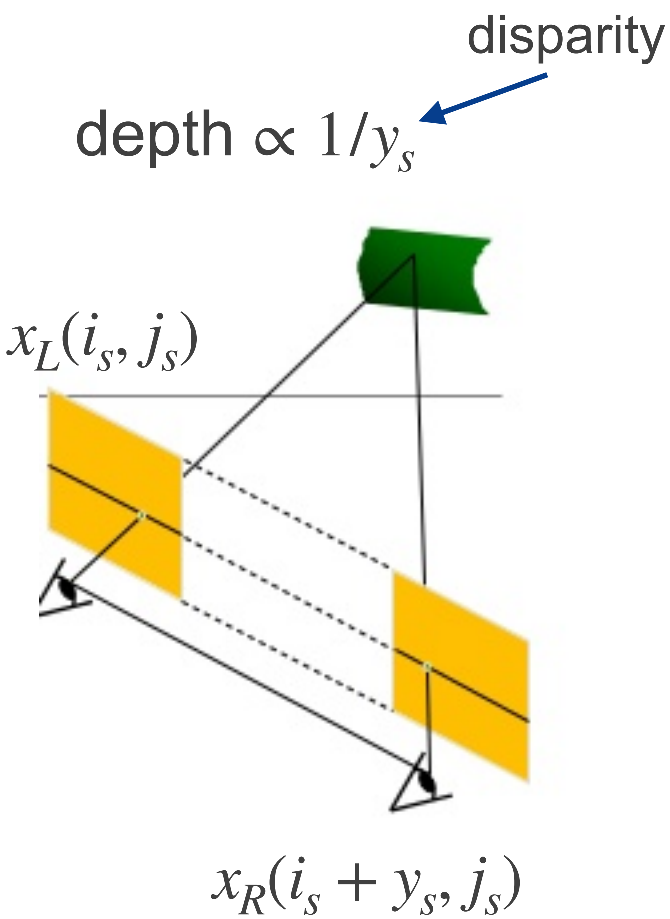
Stereo vision: determining depth of each pixel from two images taken from slightly different two angels



https://www.reallusion.com/iclone/help/iclone4/pro/09_3D_Vision/The_Concepts_of_Stereo_Vision.htm

X: two images

Y: disparities for each pixel



https://www.slideshare.net/yuhuang/passive-stereo-vision-with-deep-learning

Conditional Random Fields (CRFs)

Example: Stereo vision

$$p(\mathbf{y} \mid \mathbf{x}, \theta) = \frac{1}{Z(\mathbf{x}, \theta)} \prod_{t=1}^{T} \psi(y_t \mid \mathbf{x}, \theta) \prod_{t=1}^{T-1} \psi(y_t, y_{t+1} \mid \mathbf{x}, \theta)$$

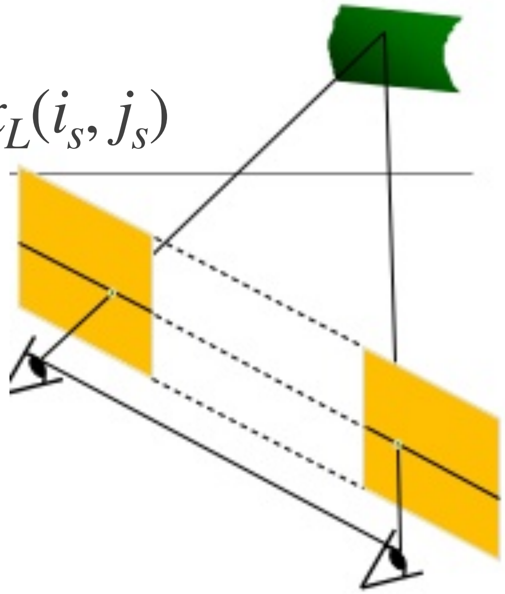
depth $\propto y_s$

Corresponding pixels have similar intensity
local node potential:

$$\psi_s(y_s \mid \mathbf{x}) \propto \exp\left[-\frac{1}{2\sigma^2}(x_L(i_s, j_s) - x_R(i_s + y_s, j_s))^2\right]$$

neighbouring disparities are similar:

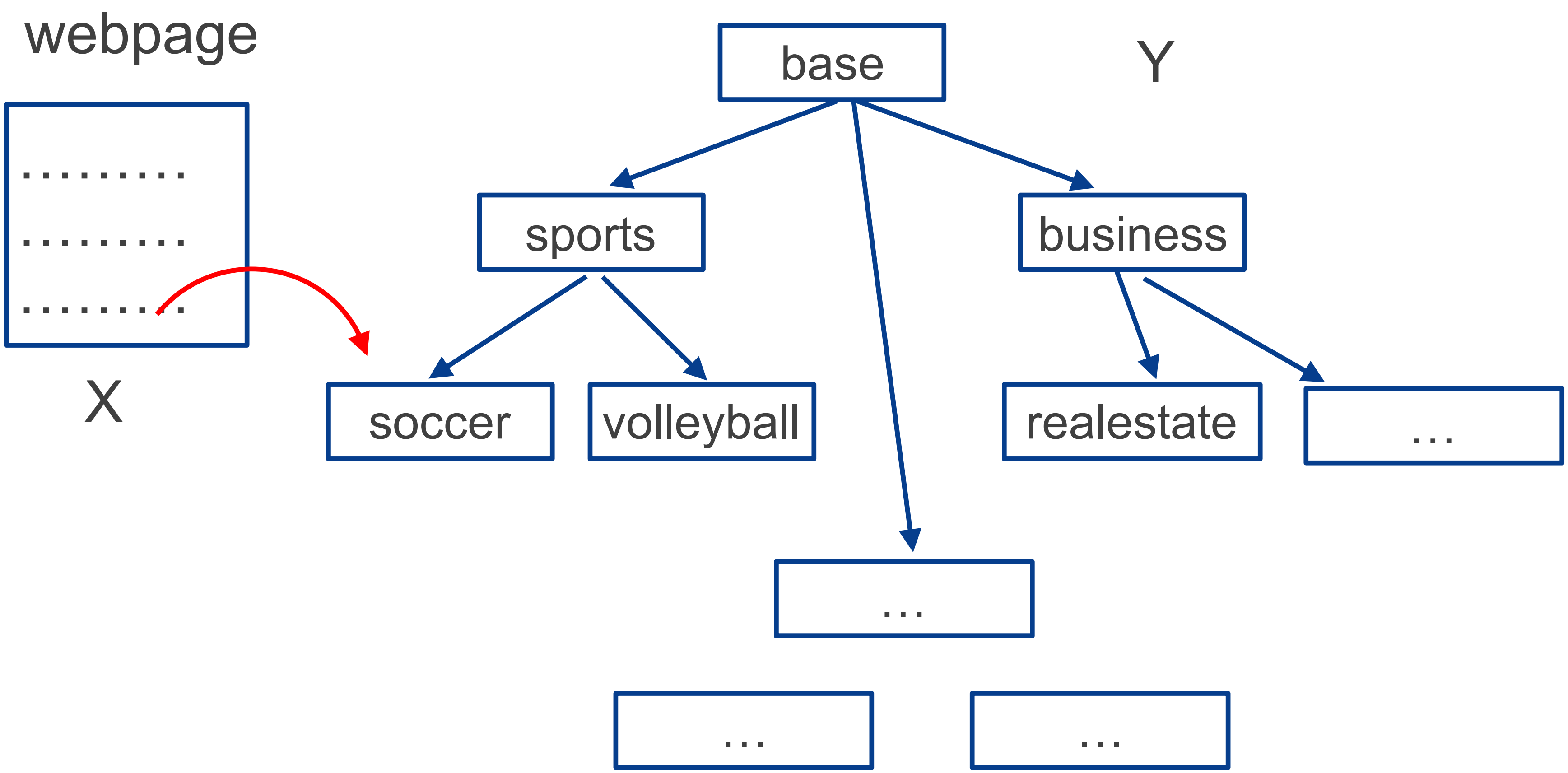
$$\psi_{st}(y_s, y_t) \propto \left[-\frac{1}{2\gamma^2}(y_s - y_t)^2\right]$$



https://www.slideshare.net/yuhuang/passive-stereo-vision-with-deep-learning

Conditional Random Fields (CRFs)
Example: Hierarchical classification

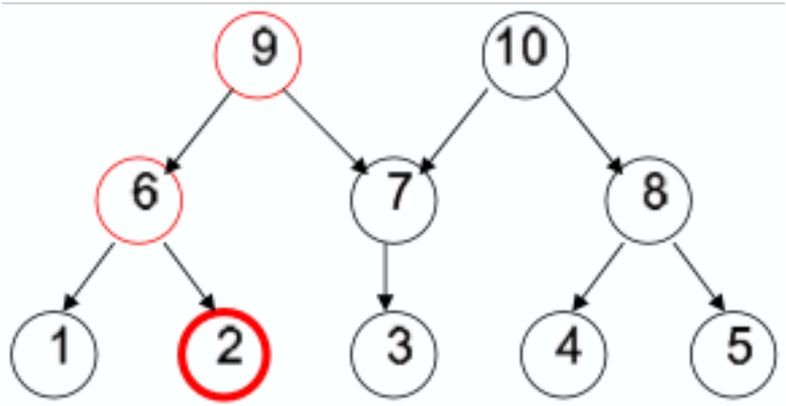
- Multi-class classification with a label taxonomy which groups the classes into a hierarchy
  - For example, web pages



Conditional Random Fields (CRFs)

Example: Hierarchical classification

- Multi-class classification with a label taxonomy which groups the classes into a hierarchy
  - For example, web pages
- Information shared between parameters for nearby categories



$$\phi(y=2) = (0,1,0,0,0,1,0,0,1,0)^T$$

$$\boldsymbol{\phi}(\mathbf{x}) = \mathbf{x}$$

$$\boldsymbol{\phi}(\mathbf{x}, y) = \phi(\mathbf{x}) \otimes \phi(y)$$

$$\phi(\mathbf{x}, y) = (0,\mathbf{x},0,0,0,\mathbf{x},0,0,\mathbf{x},0)^T$$

$$\boldsymbol{\theta}^T \boldsymbol{\phi}(\mathbf{x}, y) = \boldsymbol{\theta}_2^T \mathbf{x} + \boldsymbol{\theta}_6^T \mathbf{x} + \boldsymbol{\theta}_9^T \mathbf{x}$$

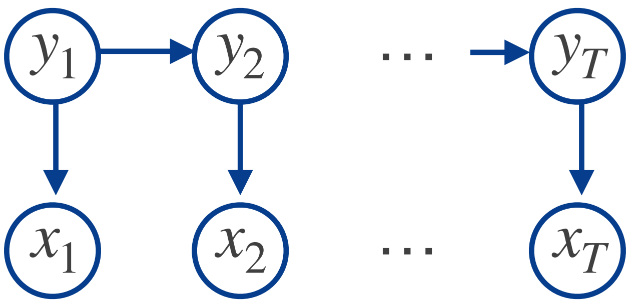
Generative vs Discriminative Models

X: 

Y: b r a c e

Generative

HMM

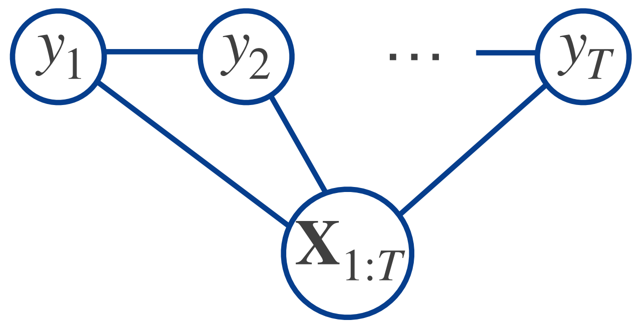


Model p(**x**,**y**)

$$p(\mathbf{x}_{1:T},\mathbf{y}_{1:T}) = p(\mathbf{y}_1)p(\mathbf{x}_1\,|\,\mathbf{y}_1)\prod_{t=2}^{T} p(\mathbf{z}_t\,|\,\mathbf{y}_{t-1})p(\mathbf{x}_t\,|\,\mathbf{y}_t)$$

Discriminative

Linear-chain CRF

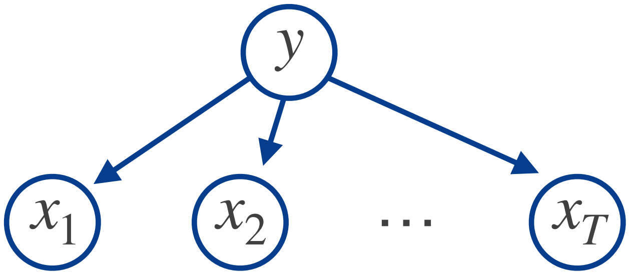


Model p(**y**|**x**) directly

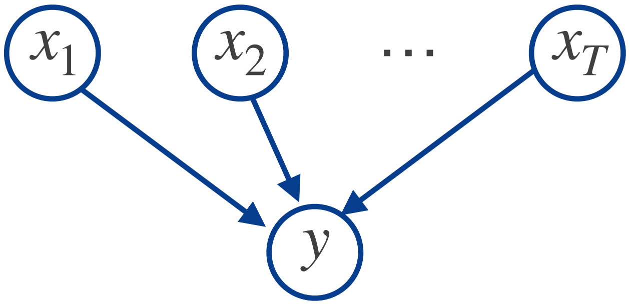
$$p(\mathbf{y}_{1:T}\,|\,\mathbf{x}_{1:T},\theta) = \frac{1}{Z(\mathbf{x},\theta)}\prod_{t=1}^{T}\psi(y_t\,|\,\mathbf{x}_{1:T},\theta)\prod_{t=1}^{T-1}\psi(y_t,y_{t+1}\,|\,\mathbf{x},\theta)$$

Generative vs Discriminative Models

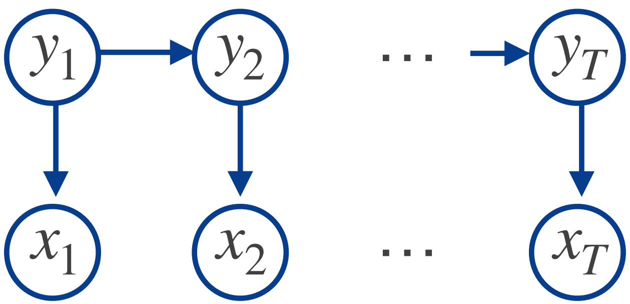
Naïve Bayes



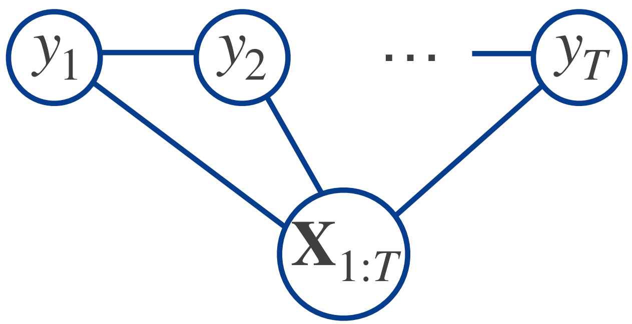
Logistic Regression



HMM



Linear-chain CRF



Model p(**x**,**y**)

Model p(**y**|**x**) directly

Advantages/Disadvantages of CRFs

- More effective in discriminative tasks
  - Focus on modelling p(**y** | **x**)
  - We can remain agnostic about p(**x**)
- Can incorporate global features easily
  - For example in NLP tasks, latent labels can depend on global properties of the sentence (e.g. language)

- They require labelled training data
- Much slower to training

CRF Training (1)

Straightforward modification of MRF training:

$$\mathcal{L}(\boldsymbol{\theta}) \stackrel{\text{def}}{=} \frac{1}{N}\sum_{n=1}^{N} \log p(\mathbf{y}^{(n)}|\mathbf{x}^{(n)}, \boldsymbol{\theta})$$

$$= \frac{1}{N}\sum_{n=1}^{N}\left[\sum_{c} \boldsymbol{\theta}_c^T \boldsymbol{\phi}_c(\mathbf{y}^{(n)}, \mathbf{x}^{(n)}) - \log Z(\boldsymbol{\theta}, \mathbf{x}^{(n)})\right]$$

- As in the MRF case:
  - Concave in θ
  - Unique global maximum using gradient-based optimisation
- Regularisation on θ usually required

CRF Training (2)

The gradients become:

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}_c} = \frac{1}{N}\sum_{n=1}^{N}\left[\phi_c(\mathbf{y}^{(n)}, \mathbf{x}^{(n)}) - \frac{\partial}{\partial \boldsymbol{\theta}_c}\log Z(\boldsymbol{\theta}, \mathbf{x}^{(n)})\right]$$

$$= \frac{1}{N}\sum_{n=1}^{N}\left\{\phi_c(\mathbf{y}^{(n)}, \mathbf{x}^{(n)}) - \boxed{\mathbb{E}_{p(\mathbf{y}|\boldsymbol{\theta}, \mathbf{x}^{(n)})}\left[\phi_c(\mathbf{y}, \mathbf{x}^{(n)})\right]}\right\}$$

- Unlike MRFs, inference is now required for every single training case inside each gradient step
  - As now the partition function depends on the inputs $\mathbf{x}^{(n)}$
  - O(N) times slower than MRFs

Training of Linear-Chain CRFs (1)

- Recall that for a linear-chain CRF we have that:

$$p(\mathbf{y}|\mathbf{x},\boldsymbol{\theta}) = \frac{1}{Z(\mathbf{x},\boldsymbol{\theta})}\prod_{t=1}^{T}\psi_t(y_t, y_{t-1}|\mathbf{x},\boldsymbol{\theta})$$

- Two inference problems arising from parameter learning
  - Log partition function $\log Z(\boldsymbol{\theta}, \mathbf{x}^{(n)})$
  - Pairwise marginal: $p(y_{t-1}, y_t|\mathbf{x}^{(n)}, \boldsymbol{\theta})$

- We would also like to predict $\mathbf{y}^\star = \underset{\mathbf{y}}{\operatorname{argmax}}\, p(\mathbf{y}|\mathbf{x}^\star, \boldsymbol{\theta})$

- We can use the same method that we developed for HMMs
- Total cost is $\mathcal{O}(TK^2NG)$:
  - N: # observations, T: sequence length, K: # states, G: # gradient comp.

Training of Linear-Chain CRFs (2)

- Recall the recursions:

$$\alpha(z_t) = p(x_t|z_t) \sum_{z_{t-1}} p(z_t|z_{t-1})\alpha(z_{t-1})$$

$$\beta(z_{t-1}) = \sum_{z} p(x_t|z_t)p(z_t|z_{t-1})\beta(z_t)$$

$$\mu(z_{t-1}) = \max_{z_t} p(x_t|z_t)p(z_t|z_{t-1})\mu(z_t)$$

- Replace $p(x_t|z_t)p(z_t|z_{t-1})$ with $\psi(y_t, y_{t-1} \mid \mathbf{x}, \theta)$ and calculate:

$$Z(\mathbf{x} \mid \theta) = \sum_{y_T} \alpha_T(y_T)$$

$$p(y_t \mid \mathbf{x}) = \frac{\alpha_t(y_t)\beta_t(y)}{Z(\mathbf{x})}$$

$$p(y_{t-1}, y_t \mid \mathbf{x}) = \frac{\alpha_{t-1}(y_{t-1})\psi(y_t, y_{t-1} \mid \mathbf{x}, \theta)\beta_t(y_t)}{Z(\mathbf{x})}$$

See Sutton, C., & McCallum, A. (2012). An introduction to conditional random fields.

Advantages and Disadvantages of UGMs

Advantages:

- They are symmetric, more natural for some domains such as spatial or relational data
- Discriminative UGMs (CRFs) work better than discriminative DGMs

Disadvantages:

- Parameters are less interpretable and less modular
- Parameter estimations is computationally more expensive

Conclusions

- UGMs motivated by image analysis and spatial statistics
  - Conditional independence properties base on “neighbouring” nodes
  - They can represent different distributions to those modelled with DGMs
- Joint distribution proportional to product of potentials
  - Local potentials do not have probabilistic interpretation
- Parameter learning via gradient-based optimisation
  - Harder than DGMs due to log partition function
- CRFs are simply UGMs conditioned on features
  - Discriminative counterpart of MRFs and HMMs
- Many applications in computer vision and NLP
- Reading: Murphy (MLaPP, 2012) Ch 19 (except Sec 19.7)